

Lecture Notes on Differential Equations and Transform Techniques

Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 19****Topics covered :**

1. Solution of ODEs for periodic inputs
2. Introduction to Fourier series

- Consider the linear ordinary differential equation (ODE)

$$a_n(x)y^{(n)} + \cdots + a_1(x)y' + a_0(x)y = g(x) \quad (1)$$

which is denoted as $L[y] = g(x)$.

Let $g_1(x)$ and $g_2(x)$ represent two functions such that $g(x) = g_1(x) + g_2(x)$. Furthermore, let $y_1(x)$ and $y_2(x)$ denote the respective solutions to the ODEs $L[y] = g_1(x)$ and $L[y] = g_2(x)$. If $y(x)$ represents the solution to the ODE $L[y] = g(x)$, then by the principle of linearity, we have $y(x) = y_1(x) + y_2(x)$. This [superposition](#) property of linear ODEs can be employed to determine the solution of an ODE when $g(x)$ is a periodic function.

- The solution to a linear ODE $L[y] = g(x)$, **when $g(x)$ is periodic**, can be obtained by combining the concepts of [Fourier series representation](#) and superposition. The process for determining the solution involves the following steps:

1. **Represent $g(x)$ as sum of sinusoidal functions:** If $g(x)$ is a periodic function with period L , it can be expressed as

$$g(x) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 x) + b_n \sin(n\omega_0 x)] \quad (2)$$

where $\omega_0 = \frac{2\pi}{L}$, where L is the period. Equation 2 is referred to as the [Fourier Synthesis Equation](#) in Trigonometric form and the coefficients a_n and b_n are referred to as the Fourier coefficients. It is important to note that while this representation is valid for many periodic functions, there are specific conditions under which all periodic functions can be expressed in this form, which we will explore.

2. **Compute the solution to generic sinusoidal functions:** The next step involves solving the ODE for specific sinusoidal functions. This means solving the following individual ODEs $L[y] = a_0$ (with solution $y_0(x)$), $L[y] = \cos(n\omega_0 x)$ (with solution $y_{c,n}(x)$) and $L[y] = \sin(n\omega_0 x)$ (with solution $y_{s,n}(x)$).
3. **Compute the overall solution:** By the principle of superposition, the solution to the ODE $L[y] = g(x)$ can be written as

$$y(x) = a_0 + \sum_{n=1}^{\infty} [a_n y_{c,n} + b_n y_{s,n}] \quad (3)$$

This representation allows for the construction of the complete solution based on the individual responses to the sinusoidal components of $g(x)$.

- Our immediate attention will be directed towards the Fourier series representation of functions and signals. Furthermore, we will use the independent variables x and t interchangeably: x will generally be used in the context of functions, while t will be used when discussing signals.
- A signal $g(t)$ is said to be **periodic**, if there exists a $T \in \mathbb{R}^{++}$ such that

$$g(t + T) = g(t) \quad (4)$$

for all $t \in \mathbb{R}$. The smallest value of T for which (4) holds is called the **fundamental period** of the signal/function. Common examples of periodic signals encountered in electrical engineering include sine and cosine signals, square waves, triangular waves, and sawtooth waves. Of these, sine and cosine functions are particularly notable for their continuity and differentiability at every point within their domain. This makes them fundamental to the Fourier Series, as they provide a smooth, well-defined basis for representing more complex periodic signals.

- The Fourier series seeks to represent a signal as a linear combination of sinusoidal functions—specifically, sine and cosine functions—whose frequencies are integer multiples of a fundamental frequency. More precisely, if $g(t)$ is a function with fundamental period T , the Fourier series attempts to express it in the following form

$$g(t) = a_0 + \sum_{i=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)] \quad (5)$$

where $\omega_0 = \frac{2\pi}{T}$. Equation (5) is known as the **Fourier Synthesis Equation**, which synthesizes a periodic function by summing sinusoidal components of different frequencies. The coefficients a_n and b_n referred to as the **Fourier coefficients**, represent the amplitudes of the cosine and sine components, respectively, at each harmonic frequency of the fundamental period. Naturally, two important questions arise

- Is it guaranteed that a given periodic function can always be expressed in the form of equation (5). This leads to the concept of convergence of the Fourier series.
- If so, what are the values of the Fourier coefficients?.

To proceed, we will first focus on deriving expressions for the Fourier coefficients. Once we have a clear understanding of how these coefficients are determined, we will then address the convergence of the Fourier series and explore the conditions under which it accurately represents a given periodic function.

- While there are various methods for deriving the Fourier coefficients and understanding Fourier analysis, a more common approach is to view it in terms of a **vector space of functions**, where the basis functions are the sinusoids $\sin(n\omega_0 x)$ and $\cos(n\omega_0 x)$ with n being an integer. We will leverage insights from the theory of vector spaces, particularly concepts from linear algebra in $\mathbb{R}^n$, to introduce key ideas associated with Fourier analysis.
- A more elegant approach to performing Fourier analysis involves the use of the complex exponential. The **complex exponential Fourier series** aims to represent a signal as a linear combination of complex exponentials, specifically

$$g(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t} \quad (6)$$

An interesting aspect of the complex exponential Fourier series is the emergence of the concept of negative frequencies. This is crucial for representing real signals in terms of complex exponentials. In particular, the terms i.e., $e^{jn\omega_0 t}$ and $e^{-jn\omega_0 t}$ combine to form real-valued sinusoidal and cosinusoidal functions.

- To better understand the role of negative frequencies, consider the analogy of an object undergoing uniform circular motion with frequency ω_0 . The position of the object in two dimensions is described by the complex exponential $e^{j\omega_0 t}$. A positive frequency $\omega_0 > 0$ corresponds to counterclockwise motion, while a negative frequency $\omega_0 < 0$ indicates clockwise motion. This analogy helps to conceptualize the idea that negative frequencies correspond to the reversal of direction in periodic motion.